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Formation of Micron-sized Presolar Graphite Grains in Low-energy Supernova Ejecta

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Abstract

Presolar graphite grains with supernova isotopic signatures provide direct evidence that large graphite grains formed in supernova ejecta. However, theoretical calculations of dust formation in supernovae have typically predicted graphite grains smaller than $0.1\,\mu\mathrm{m}$, leaving unclear the physical conditions required to form those grains with radii of order $1\,\mu\mathrm{m}$. We systematically investigate graphite nucleation and growth in expanding supernova ejecta by varying the explosion energy, ejecta mass, initial ejecta radius, and ejected ^{56}Ni mass. We use a simplified one-zone model that follows the temperature and density evolution including adiabatic cooling, radiative cooling, and heating by radioactive nickel. We show that the final graphite grain size is mainly determined by the product of the mass density of atomic carbon available for grain growth and the local cooling timescale, both evaluated at a reference temperature of $2000\,\mathrm{K}$. Our models show that graphite growth is favored by lower explosion energy, larger ejecta mass, and smaller initial ejecta radius, which keep the gas denser near the graphite-condensation temperature. When radioactive heating and radiative cooling are included, explosions with $E_{\mathrm{exp}} \sim 10^{50}\,\mathrm{erg}$ and $M_{\mathrm{Ni}} \lesssim 10^{-3}\,M_{\odot}$ can form 1–$10\,\mu\mathrm{m}$ graphite grains in the ejected helium layer or carbon–oxygen core, whereas $E_{\mathrm{exp}} \sim 10^{51}\,\mathrm{erg}$ models do not. These results suggest that micron-sized presolar supernova graphite preferentially forms in faint-supernova-like explosions with low explosion energies, little ejected nickel, and helium layer or carbon–oxygen core ejecta.

Keywords: dust, extinction — methods: numerical — supernovae: general

1 Introduction

In supernovae, newly formed or pre-existing dust can produce infrared excesses, optical attenuation and reddening, and asymmetric emission-line profiles (Matsuura et al. 2011; Gall et al. 2014; Wesson et al. 2015; Bevan & Barlow 2016; Sarangi et al. 2018; Davis et al. 2026). Such dust signatures play an important role in interpreting their light curves and spectral energy distributions. The observational impact of dust depends not only on the total dust mass but also sensitively on the grain size distribution. The grain size can affect the absorption and scattering properties, particularly when the grain size is comparable to or larger than the relevant wavelength.

Grain size also affects the survival of dust against reverse shocks and subsequent shocks driven by other supernovae. Indeed, sputtering calculations for reverse shocks have shown that small grains with $a_{\mathrm{ini}} \lesssim 0.05$–$0.1\,\mu\mathrm{m}$ are efficiently destroyed, whereas grains with $a_{\mathrm{ini}} \gtrsim 0.1$–$0.2\,\mu\mathrm{m}$ are more likely to survive (Nozawa et al. 2007; Silvia et al. 2010). In addition to this size dependence, the survival fraction in actual supernova remnants also depends on the initial grain-size distribution, dust composition, ejecta morphology, and ambient environment (Micelotta et al. 2016).

Constraining grain-size distributions from astronomical observations is challenging because the inferred distributions are generally model dependent and can be degenerate with other dust and radiative-transfer parameters (Zubko et al. 2004; Wesson et al. 2015). In contrast, laboratory measurements of individual presolar grains preserved in meteorites directly determine their sizes, compositions, and microstructures, while their isotopic ratios provide clues to their astrophysical origins (Zinner 2014; Nittler & Ciesla 2016). Among these diagnostics, excesses of $^{44}\mathrm{Ca}$ produced by the decay of $^{44}\mathrm{Ti}$ provide a representative signature of a supernova origin and have been reported for graphite grains and SiC grains from the Murchison meteorite (Nittler et al. 1996; Nittler & Ciesla 2016). Presolar grains have been identified in a variety of mineral phases. Examples include silicates such as $\mathrm{Mg_2SiO_4}$ and $\mathrm{MgSiO_3}$ (e.g., Nguyen et al. 2010), as well as alumina ($\mathrm{Al_2O_3}$) (e.g., Stroud et al. 2004). Here, we focus on graphite (C), whose single-element composition provides a simple case for modeling nucleation and grain growth. Presolar graphite grains identified as supernova condensates can reach sizes of $\sim 1\,\mu\mathrm{m}$ or larger (Amari et al. 1994; Amari et al. 2014; Nittler & Ciesla 2016), providing an important benchmark for graphite formation in supernova ejecta.

However, existing theoretical calculations of dust formation in supernova ejecta typically predict much smaller graphite grains. The characteristic sizes of graphite grains are of order several tens of nanometers, with the largest grains generally being smaller than $\sim 0.1\,\mu\mathrm{m}$ (Todini & Ferrara 2001; Nozawa et al. 2003; Sarangi & Cherchneff 2015). These grain sizes are substantially smaller than those of presolar graphite grains.

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This discrepancy may arise from the limited range of supernova properties in previous calculations. Many previous calculations of dust formation in core-collapse supernovae have adopted ejecta models representative of ordinary type II supernovae, characterized by explosion energies around $10^{51}\,\mathrm{erg}$, ejecta masses around $10\,M_\odot$, and synthesized ^{56}Ni masses of 0.01–$0.1\,M_\odot$ (Todini & Ferrara 2001; Nozawa et al. 2003; Sarangi & Cherchneff 2015). However, supernovae span a broad range of explosion energies, ejecta masses, and synthesized ^{56}Ni masses. Even among ordinary core-collapse supernovae, typical inferred ranges are explosion energies of $\sim 10^{50}$–$10^{51}\,\mathrm{erg}$, ejecta masses of $\sim$ a few–$10M_\odot$, and synthesized ^{56}Ni masses of $\sim 10^{-3}$–$10^{-1}M_\odot$ (Müller et al. 2017; Martinez et al. 2022b). At the lower-energy end, faint supernovae and failed-supernova candidates extend this diversity toward explosions with very small ^{56}Ni production and weak mass ejection (Hamuy 2003; Pejcha & Prieto 2015; Adams et al. 2017; Taddia et al. 2018; Martinez et al. 2022b; De et al. 2026a; De et al. 2026b). These macroscopic parameters affect the temperature and density evolution of the ejecta through expansion, density dilution, radioactive heating, and radiative cooling, and therefore directly influence the conditions under which dust forms and grows. Despite this diversity, how the final grain size changes over a broader parameter space, particularly toward lower explosion energies and smaller ejected ^{56}Ni masses, has not been systematically explored.

In this paper, we aim to study graphite dust formation under various conditions for supernova explosion. To this end, we develop a simple one-zone model to treat dust formation in supernova ejecta to facilitate an efficient parameter study. Our main goal is to clarify how the explosion energy, ejecta mass, ejected ^{56}Ni mass, and ejecta layer affect the final graphite grain size through the thermal history of the ejecta. We find that micron-sized graphite preferentially forms in low-energy, nickel-poor explosions with relatively large ejecta masses and ejected helium layer or carbon–oxygen core material, where the gas remains sufficiently dense near the graphite-condensation temperature.

This paper is organized as follows. In Section 2, we describe the one-zone ejecta model and the treatment of its thermal evolution. In Section 3, we present the calculation of graphite nucleation and grain growth based on classical nucleation theory. In Section 4, we examine the conditions under which micron-sized graphite can form, using both Ni-free benchmark models and a parameter survey including ^{56}Ni heating. In Section 5, we discuss the implications of our results for low-energy supernovae and the origin of presolar supernova graphite. Finally, we summarize our conclusions in Section 6.

2 One-zone model of supernova ejecta

To calculate graphite nucleation and grain growth in supernova ejecta, we first consider the time evolution of the gas density and temperature. The resulting density–temperature trajectories determine the supersaturation ratio, the nucleation rate, and the subsequent growth rate of graphite grains, and are used as input for the dust-formation calculation described in Section 3.

In this study, we use a simplified one-zone ejecta model. We assume that the ejecta is spherical and spatially homogeneous, with volume $V = 4\pi R^3/3$ and density $\rho = 3M_{\rm ej}/(4\pi R^3)$, where R is the characteristic ejecta radius. The model is specified by the initial explosion energy $E_{\rm exp}$, ejecta mass $M_{\rm ej}$, initial ^{56}Ni mass $M_{\rm Ni,0}$, initial radius R_0, and initial internal-energy fraction $f_{\rm int}$. We define $t = 0$ as the initial epoch of the one-zone evolution and $E_{\rm exp} \equiv E_{\rm int}(0) + E_{\rm kin}(0)$, where $E_{\rm int}$ and $E_{\rm kin}$ are the internal and kinetic energies of the ejecta, respectively. The initial conditions are $R(0) = R_0,\ E_{\rm int}(0) = f_{\rm int}E_{\rm exp},\ E_{\rm kin}(0) = (1 - f_{\rm int})E_{\rm exp}$.

At a given radius R, the internal energy determines the temperature and pressure through an equation of state that includes radiation and ideal-gas contributions,

$$\varepsilon = aT^4 + \frac{3}{2}\frac{\rho}{\bar{\mu}m_p}k_BT, \tag{1}$$

where $\varepsilon \equiv E_{\rm int}/V$, a, m_p, k_B, $\bar{\mu}$ are the internal energy density, radiation constant, proton mass, Boltzmann constant, and mean molecular weight defined by the total number density of ions and electrons, respectively. The pressure is then calculated as

$$p = \frac{aT^4}{3} + \frac{\rho}{\bar{\mu}m_p}k_BT. \tag{2}$$

We adopt a fixed mean molecular weight of $\bar{\mu} = 1$ in all calculations.

The thermal evolution of the ejecta is calculated by solving the energy equation including radioactive heating, radiative cooling, and adiabatic expansion (Arnett 1980; Arnett 1982). The internal energy evolves as

$$\frac{dE_{\rm int}}{dt} = -L + \dot{Q} - p\frac{dV}{dt}, \tag{3}$$

where L and $\dot{Q}$ are the radiative luminosity and radioactive heating rate.

We first describe the radiative luminosity L in Eq. (3). In the optically thick limit, the corresponding diffusion luminosity is written as

$$L_{\rm diff} = \frac{E_{\rm rad}}{t_{\rm diff}}, \tag{4}$$

where $E_{\rm rad} = aT^4V$ is the radiation energy and $t_{\rm diff}$ is the diffusion timescale. For the one-zone ejecta, the diffusion timescale is

$$t_{\rm diff} = \frac{\tau R}{c}, \tag{5}$$

where c is the speed of light and τ is the optical depth,

$$\tau = \kappa\rho R = \frac{3\kappa M_{\rm ej}}{4\pi R^2}. \tag{6}$$

Here, κ denotes the Rosseland-mean opacity. In the optically thin limit, photons escape on the timescale

$$t_{\rm lc} = \frac{R}{c}, \tag{7}$$

where $t_{\rm lc}$ is the light-crossing timescale. We interpolate between the optically thick and thin limits by taking the effective photon escape timescale to be $t_{\rm diff} + t_{\rm lc}$. The radiative luminosity is then

$$L = \frac{E_{\rm rad}}{t_{\rm diff} + t_{\rm lc}} = \frac{\tau}{1+\tau}L_{\rm diff}, \tag{8}$$

which approaches $L = L_{\rm diff}$ in the optically thick limit and $L = E_{\rm rad}/t_{\rm lc}$ in the optically thin limit (Metzger et al. 2014; Metzger & Pejcha 2017; Chen & Liang 2024). In this work, we adopt a constant electron-scattering opacity, $\kappa = 0.2\,\mathrm{cm^2\,g^{-1}}$ for all models with radiative cooling, as commonly adopted in simplified supernova diffusion models (Kasen & Bildsten 2010; Metzger et al. 2014; Metzger et al. 2015). For comparison, we also consider a purely adiabatic model by setting $L = 0$, which isolates the effect of pdV work and radioactive heating on the thermal evolution of the ejecta.

We next describe the radioactive heating term $\dot{Q}$ in Eq. (3). The $^{56}\mathrm{Ni} \rightarrow {}^{56}\mathrm{Co} \rightarrow {}^{56}\mathrm{Fe}$ decay chain releases energy in γ-rays, positrons, and neutrinos (Nadyozhin 1994). We neglect neutrino

energy deposition and distinguish between the γ-ray and positron kinetic-energy contributions to the deposited heating rate:

$$\dot{Q} = f_{\rm dep}(\tau_\gamma)\,\dot{Q}_\gamma + \dot{Q}_{\rm pos}. \tag{9}$$

We adopt the radioactive energy-release rates for the ^{56}Ni $\rightarrow$ ^{56}Co $\rightarrow$ ^{56}Fe decay chain given by Sharon & Kushnir (2020):

$$\dot{Q}_\gamma(t) = \frac{M_{\rm Ni,0}}{M_\odot}\left[6.54\,e^{-t/\tau_{\rm Ni}} + 1.38\,e^{-t/\tau_{\rm Co}}\right] \times 10^{43}\,{\rm erg\,s^{-1}},$$
$$\dot{Q}_{\rm pos}(t) = 4.64\,\frac{M_{\rm Ni,0}}{M_\odot}\left[e^{-t/\tau_{\rm Co}} - e^{-t/\tau_{\rm Ni}}\right] \times 10^{41}\,{\rm erg\,s^{-1}}, \tag{10}$$

where $M_{\rm Ni,0}$ is the initial ^{56}Ni mass, and $\tau_{\rm Ni} = 8.76\,{\rm d}$ and $\tau_{\rm Co} = 111.4\,{\rm d}$ are the mean lifetimes. The γ-ray component includes photons produced by positron annihilation, whereas $\dot{Q}_{\rm pos}$ contains only the positron kinetic energy. We assume that this kinetic energy is deposited locally and instantaneously. To describe incomplete γ-ray trapping in our one-zone model, we approximate the deposited fraction as

$$f_{\rm dep}(\tau_\gamma) = 1 - e^{-\tau_\gamma}. \tag{11}$$

We parameterize the γ-ray optical depth as

$$\tau_\gamma = \kappa_\gamma \rho R = \frac{3\kappa_\gamma M_{\rm ej}}{4\pi R^2}, \tag{12}$$

where κ_γ is the effective (gray) γ-ray opacity. We adopt a constant value of $\kappa_\gamma = 0.027\,{\rm cm^2\,g^{-1}}$, which is reasonable in the γ-ray transport in supernova ejecta (Sutherland & Wheeler 1984; Maeda et al. 2003).

Finally, we describe the dynamical evolution associated with the pdV work in Eq. (3). The kinetic energy evolves as

$$\frac{dE_{\rm kin}}{dt} = p\frac{dV}{dt}, \tag{13}$$

where $E_{\rm kin}$ is the kinetic energy of the ejecta. The ejecta radius then evolves as

$$\frac{dR}{dt} = \left(\frac{2E_{\rm kin}}{M_{\rm ej}}\right)^{\frac{1}{2}}. \tag{14}$$

We simultaneously solve Eq. (3), Eq. (13), and Eq. (14) as three coupled ordinary differential equations for $E_{\rm int}$, $E_{\rm kin}$, and R.

Fig. 1 shows the resulting density–temperature trajectories for representative combinations of explosion parameters. At fixed $M_{\rm ej}$, R_0, and $f_{\rm int}$, increasing $E_{\rm exp}$ from 10^{48} to $10^{51}\,{\rm erg}$ raises the initial temperature and shifts the trajectory toward higher temperatures at a given density. Adding $M_{\rm Ni,0} = 0.05\,M_\odot$ keeps the $10^{48}\,{\rm erg}$ ejecta hotter during expansion, so they reach temperatures near condensation ($\sim 1000\,{\rm K}$) at lower densities. The effect of radioactive heating is much smaller for $E_{\rm exp} = 10^{51}\,{\rm erg}$.

The gray line indicates where the radiation and ideal-gas contributions to the internal energy in Eq. (1) are equal; the radiation contribution dominates above this line, whereas the ideal-gas contribution dominates below it. For reference, adiabatic expansion gives $T \propto \rho^{1/3}$ in the radiation-dominated limit and $T \propto \rho^{2/3}$ in the matter-dominated limit, as indicated by the short gray line segments in Fig. 1. Some trajectories evolve from the radiation-dominated to the matter-dominated regime during expansion, whereas others are already matter dominated from the beginning of the evolution shown here.

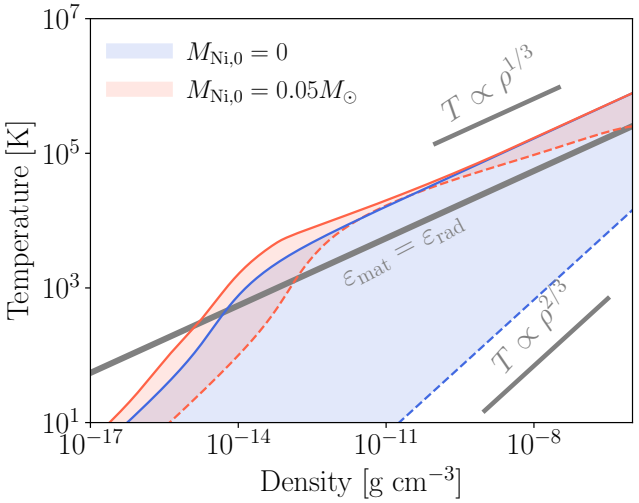


Fig. 1. The density–temperature trajectories for representative combinations of explosion energy and initial ^{56}Ni mass. Colors indicate $M_{\rm Ni,0}$, while line styles indicate $E_{\rm exp}$ (solid lines: $10^{51}\,{\rm erg}$, dashed lines: $10^{48}\,{\rm erg}$). The gray thick line indicates $\varepsilon_{\rm mat} = \varepsilon_{\rm rad}$, where $\varepsilon_{\rm rad} = aT^4$ and $\varepsilon_{\rm mat} = 3\rho k_B T/2m_p$ are the radiation and ideal-gas contributions to the internal energy density, respectively. The short gray line segments indicate the adiabatic scalings $T \propto \rho^{1/3}$ and $T \propto \rho^{2/3}$ in the radiation- and matter-dominated limits, respectively. We use $f_{\rm int} = 0.1$, $M_{\rm ej} = 10\,M_\odot$, and $R_0 = 10^{13}\,{\rm cm}$ for all models.

3 Dust Nucleation and Grain Growth

In this study, dust formation in supernova ejecta is treated as a two-step process consisting of nucleation and subsequent grain growth. We calculate these processes along the density and temperature evolution obtained in Section 2, so that the thermal history of the ejecta directly determines the conditions for dust nucleation and grain growth. The basic framework of the dust formation calculation follows Nozawa et al. (2003), and is formulated using steady-state classical nucleation theory under the key species approximation.

3.1 Overview of nucleation theory

In cooling and expanding ejecta, atoms and molecules in the gas phase form condensation nuclei, which subsequently grow into dust grains. For each dust species j, the nucleation rate is described by the steady-state nucleation rate J_j. Following Nozawa et al. (2003), we formulate nucleation and grain growth using the key species approximation introduced by Kozasa & Hasegawa (1987). Under this approximation, both the nucleation and grain-growth rates are assumed to be controlled by the least abundant colliding gas species in the reaction network. Although the formalism is written for a general dust species j, we consider only graphite in this study, for which atomic carbon is the key species.

Under this approximation, the steady-state nucleation rate for dust species j is given by

$$J_j = \alpha_j \Omega_{1j} c_{1j}^2 \sqrt{\frac{2\zeta_j}{\pi m_{1j}}} \exp\left[-\frac{4}{27}\frac{\mu_j^3}{(\ln S_j)^2}\right], \tag{15}$$

where α_j, ζ_j, and S_j denote the sticking probability, surface free energy, and supersaturation ratio of the key species for dust species j, respectively. We assume $\alpha_j = 1$ throughout this study. Here, $\Omega_{1j} \equiv 4\pi a_{1j}^3/3$ is the volume of a monomer, while c_{1j}, m_{1j}, and a_{1j} represent the number density in the gas phase, mass, and atomic radius of the key species, respectively. The dimensionless surface tension is defined as

$$\mu_j \equiv \frac{4\pi a_{1j}^2 \zeta_j}{k_B T}, \tag{16}$$

where we have assumed that the temperature of the condensed phase is equal to the gas temperature.

The supersaturation ratio S_j is defined by taking into account the chemical reactions relevant to dust formation as

$$\ln S_j = -\frac{\Delta G_j^0}{k_B T} + \sum_i \nu_{ij} \ln P_{ij}, \tag{17}$$

where ΔG_j^0 is the standard Gibbs free energy of formation per key species molecule, P_{ij} denotes the partial pressure of reactants and products, and ν_{ij} is the stoichiometric coefficient. As the ejecta cools and expands, nucleation begins once S_j exceeds unity. We set $J_j = 0$ for $S_j \leq 1$ and evaluate Eq. (15) only for $S_j > 1$.

3.2 Grain growth and depletion of key species

After nucleation, dust grains are assumed to grow by surface accretion through collisions with key species. The growth rate of the grain size r_j is proportional to the number density of the key species and is given by

$$\frac{dr_j}{dt} = \alpha_j \, \Omega_{1j} \, c_{1j} \, v_{\mathrm{th},j}, \tag{18}$$

where

$$v_{\mathrm{th},j} = \sqrt{\frac{k_B T}{2\pi m_{1j}}} \tag{19}$$

is the thermal velocity of the key species.

To account for the depletion of the key species, we introduce the normalized monomer abundance,

$$Y_j \equiv \frac{c_{1j}}{\tilde{c}_{1j}}, \tag{20}$$

where $\tilde{c}_{1j}$ is the number density of the key species in the absence of depletion by dust formation. The nucleation rate and grain growth are coupled self-consistently by solving the following integral equation:

$$1 - Y_j = \int_{t_S}^{t} \frac{1}{\tilde{c}_{1j}} J_j(t') \frac{1}{\Omega_{1j}} \frac{4\pi}{3} r_j^3(t,t') \, dt'. \tag{21}$$

Here, $r_j(t,t')$ represents the radius at time t of a grain nucleated at time t', and t_S is the time at which S_j first exceeds unity.

To describe the evolution of the grain population, we introduce the ℓ-th moment as

$$K_j^{(\ell)} = \int_{t_S}^{t} \frac{J_j(t')}{\tilde{c}_{1j}(t')\Omega_{1j}} \frac{4\pi}{3} r_j^{\ell}(t,t') \, dt'. \tag{22}$$

Differentiating Eq. (22) gives

$$\frac{dK_j^{(\ell)}}{dt} = \frac{J_j}{\tilde{c}_{1j}} \frac{4\pi}{3\Omega_{1j}} r_{*j}^{\ell} + \ell K_j^{(\ell-1)} \frac{dr_j}{dt} \qquad (\ell = 1,2,3), \tag{23}$$

where

$$r_{*j} \equiv \frac{2 a_{1j} \mu_j}{3 \ln S_j} \tag{24}$$

is the critical radius required for a stable nucleus. During graphite nucleation in our models, r_{*j} is typically of order 10^{-8}–$10^{-7}\,\mathrm{cm}$, corresponding to a few monomer radii. For the zeroth moment, Eq. (22) gives

$$\frac{dK_j^{(0)}}{dt} = \frac{J_j}{\tilde{c}_{1j}} \frac{4\pi}{3\Omega_{1j}}. \tag{25}$$

For graphite, Eqs. (21) and (22) give

$$\frac{dY_j}{dt} = -\frac{dK_j^{(3)}}{dt}, \tag{26}$$

with $Y_j(0) = 1$.

We simultaneously integrate Eqs. (18), (23), (25), and (26) for r_j, $K_j^{(0)}$–$K_j^{(3)}$, and Y_j, using $c_{1j} = Y_j \tilde{c}_{1j}$ at each time step. The mass-averaged grain size $r_{\mathrm{gr},j}$ and the grain number density $n_{\mathrm{gr},j}$ are then obtained from the moments as

$$r_{\mathrm{gr},j} = \left(\frac{K_j^{(3)}}{K_j^{(0)}} \right)^{1/3}, \tag{27}$$

and

$$n_{\mathrm{gr},j} = a_{1j}^3 K_j^{(0)} \tilde{c}_{1j}. \tag{28}$$

3.3 Computational setup and simplifications

In this study, we focus exclusively on the formation of graphite and do not consider the formation of silicate or other dust species. In chemically mixed ejecta with solar abundance ratios ($\mathrm{C/O} < 1$), carbon is expected to be predominantly locked into CO molecules, leaving little free carbon available for graphite formation.

The number density of carbon key species contributing to graphite formation is parameterized as

$$\tilde{c}_{1\mathrm{C}} = (1 - f_{\mathrm{CO}}) \frac{X_{\mathrm{C}} \rho}{m_{1\mathrm{C}}}, \tag{29}$$

where X_{C} and f_{CO} are the carbon mass fraction in the ejecta and the fraction of carbon locked into CO molecules, respectively, and $m_{1\mathrm{C}} \equiv 12 m_p$ is the carbon atom mass.

Rather than assuming complete CO formation ($f_{\mathrm{CO}} = 1$) for $\mathrm{C/O} < 1$, we treat the effective carbon abundance

$$X'_{\mathrm{C}} \equiv (1 - f_{\mathrm{CO}}) X_{\mathrm{C}} \tag{30}$$

as a free parameter. This parametrization allows us to systematically explore the impact of incomplete CO formation on graphite nucleation without explicitly modeling the non-equilibrium CO chemistry, including CO destruction by energetic particles produced by radioactive decay (Clayton et al. 1999; Cherchneff & Dwek 2009; Sarangi & Cherchneff 2013).

For graphite, we adopt the monomer radius of $a_{1\mathrm{C}} = 1.281 \times 10^{-8}\,\mathrm{cm}$, a surface free energy of $\zeta_{\mathrm{C}} = 1400\,\mathrm{erg\,cm^{-2}}$, and approximate the Gibbs free energy for carbon condensation as $\Delta G_{\mathrm{C}}^0/(k_B T) = -A/T + B$, with $A = 8.64726 \times 10^4\,\mathrm{K}$ and $B = 19.0422$, following Nozawa et al. (2003).

We use the density and temperature histories obtained in Section 2 as input to the dust-formation calculation. The density $\rho(t)$ sets the undepleted carbon number density, $\tilde{c}_{1\mathrm{C}} = X'_{\mathrm{C}} \rho / m_{1\mathrm{C}}$, while monomer depletion gives $c_{1\mathrm{C}} = Y_{\mathrm{C}} \tilde{c}_{1\mathrm{C}}$. The temperature $T(t)$ determines μ_{C} and $v_{\mathrm{th,C}}$ and, together with the carbon partial pressure, determines $\ln S_{\mathrm{C}}$. These quantities enter the nucleation and grain-growth rates through Eqs. (15) and (18), respectively.

4 Results

4.1 Time evolution and final sizes of graphite grains

To illustrate how the explosion energy and ejecta mass affect graphite nucleation and growth, we first consider Ni-free adiabatic models with $E_{\mathrm{exp}} = 10^{48}\,\mathrm{erg}$ and $10^{51}\,\mathrm{erg}$. For simplicity, we omit radioactive heating and radiative cooling here, although those effects will be considered in Section 4.4. The colors indicate the ejecta mass, which ranges from 0.3 to $30\,M_\odot$ at each explosion energy.

We characterize graphite formation by the nucleation rate J, the grain number density n_{gr}, and the mass-averaged grain radius

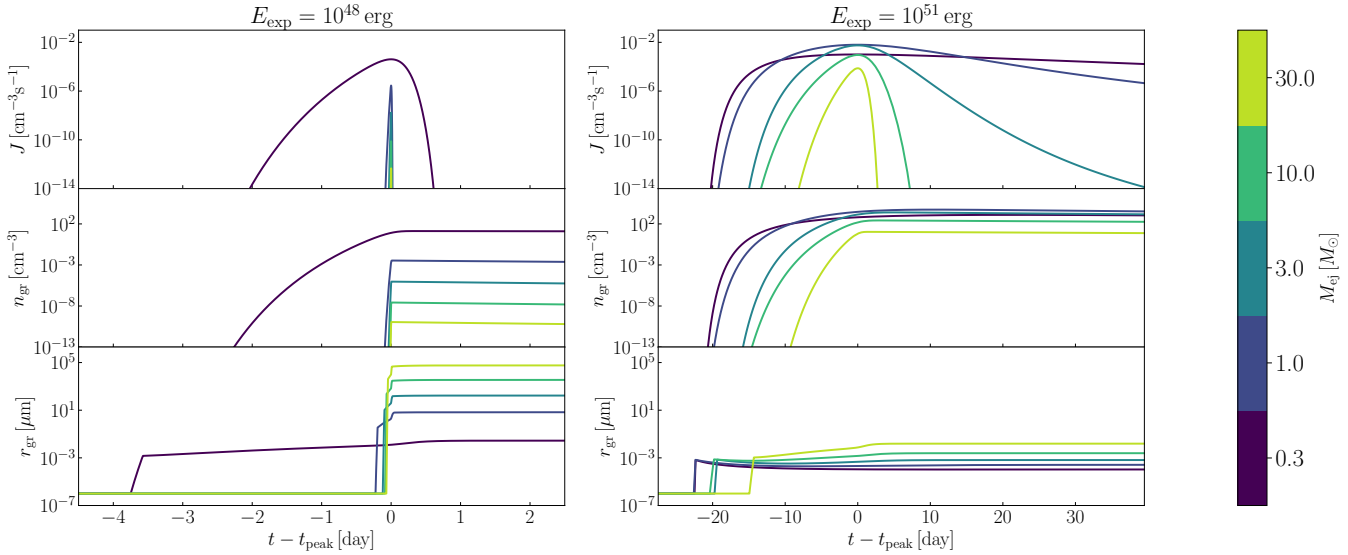


Fig. 2. Time evolution of the steady-state nucleation rate J, grain number density $n_{\rm gr}$, and mass-averaged graphite grain size $r_{\rm gr}$ for different ejecta masses $M_{\rm ej}$. The left and right columns show models with $E_{\rm exp} = 10^{48}\,{\rm erg}$ and $10^{51}\,{\rm erg}$, respectively. In each model, the time is measured relative to the peak of the nucleation rate, $t_{\rm peak}$. The top, middle, and bottom panels show J, $n_{\rm gr}$, and $r_{\rm gr}$, respectively. The color bar indicates $M_{\rm ej}$. All models use $f_{\rm int} = 0.9$, $R_0 = 10^{12}\,{\rm cm}$, and $X'_{\rm C} = 3 \times 10^{-3}$.

Table 1. Model summary for $R_0 = 10^{12}\,{\rm cm}$, $f_{\rm int} = 0.9$, and $X'_{\rm C} = 3 \times 10^{-3}$.

$E_{\rm exp}$ [erg]	$M_{\rm ej}$ [$M_\odot$]	$t_{\rm peak}$ [day]	$r_{\rm gr,final}$ [μm]	$\rho_{\rm peak}$ [g cm^{-3}]	$T_{\rm peak}$ [K]	$\tau_{\rm cool,peak}$ [day]
10^{48}	0.3	110	2.8×10^{-2}	9.4×10^{-13}	1900	180
	1	19	6.8	3.4×10^{-9}	2400	24
	3	13	170	1.4×10^{-7}	2700	13
	10	12	3.5×10^{3}	3.6×10^{-6}	3100	16
	30	11	5.8×10^{4}	6.8×10^{-5}	3500	14
10^{51}	0.3	79	1.1×10^{-4}	7.4×10^{-17}	1200	180
	1	120	2.6×10^{-4}	4.6×10^{-16}	1400	260
	3	170	6.6×10^{-4}	2.5×10^{-15}	1500	410
	10	260	2.5×10^{-3}	1.3×10^{-14}	1600	640
	30	300	1.6×10^{-2}	1.3×10^{-13}	1800	630

$r_{\rm gr}$. Fig. 2 shows their time evolution in the top, middle, and bottom panels, respectively. The left and right columns correspond to $E_{\rm exp} = 10^{48}\,{\rm erg}$ and $10^{51}\,{\rm erg}$, and the colors indicate the ejecta mass. For each model, we define $t_{\rm peak}$ as the time when J reaches its maximum and plot these quantities against $t - t_{\rm peak}$, so that all nucleation peaks are aligned at zero. Table 1 lists $t_{\rm peak}$ measured from the start of the calculation. For $E_{\rm exp} = 10^{48}\,{\rm erg}$, $t_{\rm peak}$ decreases from 110 to 11 days as $M_{\rm ej}$ increases from 0.3 to $30\,M_\odot$. For $E_{\rm exp} = 10^{51}\,{\rm erg}$, the dependence is monotonically increasing. The peak time increases from 79 to 300 days between 0.3 and $30\,M_\odot$. At fixed ejecta mass, the higher-energy models reach the nucleation peak later except for $M_{\rm ej} = 0.3 M_\odot$.

Around the nucleation peak, $n_{\rm gr}$ rises rapidly as new grains form. Subsequent accretion increases $r_{\rm gr}$ toward its final value, producing a step-like evolution. At both explosion energies, increasing the ejecta mass increases the final grain radius. When fewer grains nucleate, the available carbon monomers are accreted onto fewer grains, allowing each grain to grow more efficiently. The subsequent grain growth is rapidly suppressed by the decreasing gas density due to ejecta expansion and by the depletion of carbon monomers in the gas phase.

The temporal structure, however, differs between the two explosion energies. For $E_{\rm exp} = 10^{48}\,{\rm erg}$, nucleation and rapid grain growth are strongly concentrated around $t_{\rm peak}$. For $E_{\rm exp} = 10^{51}\,{\rm erg}$, the nucleation rate can evolve over a broader interval, particularly in the lower-mass models. Grain growth by gas accretion follows and is completed immediately after the formation of seed nuclei (characterized by J), resulting in $r_{\rm gr}$ tracks that resemble a step function.

According to Eq. (18), the higher key-species number density gives a larger grain growth rate around $t_{\rm peak}$, resulting in a larger final grain size. The properties of the models with $E_{\rm exp} = 10^{48}\,{\rm erg}$ and $10^{51}\,{\rm erg}$ are summarized in Table 1. At the same ejecta mass, the higher-energy models generally reach the nucleation peak at lower densities and produce smaller graphite grains. For example, at $M_{\rm ej} = 10 M_\odot$, increasing the explosion energy from $10^{48}\,{\rm erg}$ to $10^{51}\,{\rm erg}$ reduces the density at the nucleation peak from 3.6×10^{-6} to $1.3 \times 10^{-14}\,{\rm g\,cm^{-3}}$, and the final grain size from 3.5×10^{3} to $2.5 \times 10^{-3}\,\mu{\rm m}$.

We next examine the final grain size over a wider range of $E_{\rm exp}$, $M_{\rm ej}$, and R_0 in Ni-free adiabatic models. Fig. 3 shows the final mass-averaged graphite grain radius on the $E_{\rm exp}$–$M_{\rm ej}$ plane. Throughout these calculations, we fix $f_{\rm int} = 0.9$ and $X'_{\rm C} = 3 \times 10^{-3}$. The contours at $r_{\rm gr} = 0.01$, 1, and $100\,\mu{\rm m}$ run ap-

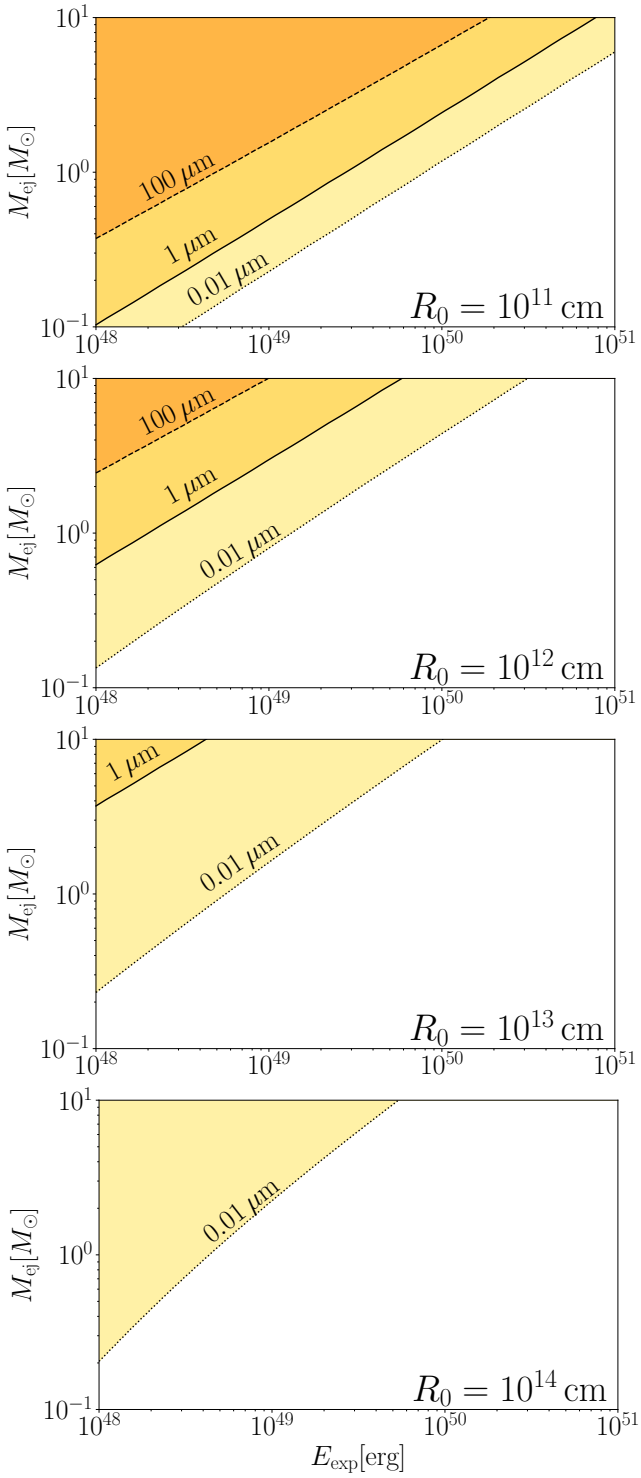


Fig. 3. Final mass-averaged dust radius on the $E_{\rm exp}$-$M_{\rm ej}$ plane for Ni-free ejecta with four values of the initial radius R_0: $R_0 = 10^{11}$, 10^{12}, 10^{13}, and 10^{14} cm, shown from top to bottom, respectively. The horizontal axis is the explosion energy $E_{\rm exp}$, and the vertical axis is the ejecta mass $M_{\rm ej}$. The dotted, solid, and dashed lines indicate $r_{\rm gr} = 0.01\,\mu$m, $1\,\mu$m, and $100\,\mu$m, respectively. The shaded regions show the corresponding ranges separated by these contours.

proximately from the lower left to the upper right as straight lines in the logarithmic $E_{\rm exp}$–$M_{\rm ej}$ plane.

At each fixed R_0, the final grain size increases with increasing $M_{\rm ej}$ and decreasing $E_{\rm exp}$. The contours of constant grain size therefore shift toward larger $M_{\rm ej}$ as $E_{\rm exp}$ increases. In particular, models above the $r_{\rm gr} = 1\,\mu$m contour produce micron-sized or larger grains, showing that higher explosion energies require larger ejecta masses to reach this size.

The initial radius also strongly affects the final grain size. As R_0 increases, the region producing micron-sized grains contracts toward lower $E_{\rm exp}$ and larger $M_{\rm ej}$. For example, at $E_{\rm exp} = 10^{49}$ erg and $M_{\rm ej} = 10\,M_\odot$, the final grain size exceeds $1\,\mu$m for $R_0 = 10^{12}$ cm but falls below this value for $R_0 = 10^{13}$ cm. For $R_0 = 10^{14}$ cm, no models reach $1\,\mu$m within the parameter range shown in Fig. 3.

4.2 Analytical estimates of the grain size

Fig. 2 shows that graphite grains grow mainly during a short interval around the peak of the steady-state nucleation rate. This suggests that, despite the different evolutionary histories, the final grain size may be characterized by the local thermodynamic conditions during the graphite-growth phase. We therefore seek a simple quantity that connects these local conditions to the final grain size across different ejecta models.

Guided by the peak nucleation temperatures in Table 1, we adopt $T = 2000$ K as a representative temperature of the graphite-growth phase. Evaluating Eq. (18) at this temperature with the undepleted carbon number density $X'_{\rm C}\rho_{2000}/m_{\rm 1C}$, we define the reference growth rate as

$$\left.\frac{dr}{dt}\right|_{\rm ref} \equiv \frac{\alpha_{\rm C}\Omega_{\rm 1C}v_{\rm th,2000}}{m_{\rm 1C}} X'_{\rm C}\rho_{2000}. \tag{31}$$

The characteristic increase in grain radius is then approximated as $\Delta r \simeq (dr/dt|_{\rm ref})\Delta t_{\rm eff}$. We parameterize the effective growth duration as

$$\Delta t_{\rm eff} = f_{\rm grow}\tau_{\rm cool,2000}. \tag{32}$$

Here, $\tau_{\rm cool,2000} \equiv |d\ln T/dt|^{-1}_{\rm 2000K}$ is the cooling timescale evaluated when the gas temperature reaches 2000 K. The dimensionless factor $f_{\rm grow}$ accounts for the finite growth interval, monomer depletion, and changes in the growth rate relative to the reference value. A more detailed derivation is given in Appendix 2. Retaining $a_{\rm 1C}$ as the monomer-size baseline, we obtain the estimate

$$r_{\rm gr,final} \simeq a_{\rm 1C} + f_{\rm grow}\frac{\alpha_{\rm C}\Omega_{\rm 1C}v_{\rm th,2000}}{m_{\rm 1C}} X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}. \tag{33}$$

Eq. (33) suggests $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ as a useful quantity for characterizing grain growth. However, $f_{\rm grow}$ can vary between models. We therefore examine numerically whether models with similar values of this product yield similar final grain sizes. Fig. 4 shows this comparison for models with $R_0 = 10^{13}$ cm. Although the explosion energy is varied, the results approximately collapse onto a single relation. We have confirmed that the same collapse is obtained for other values of R_0 in the range 10^{11}–10^{14} cm. The numerical results thus reveal an approximately single-valued relation between $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ and the final grain size over the explored range of Ni-free adiabatic models. Different combinations of explosion parameters that give similar values of this product yield similar final grain sizes.

For sufficiently large values of $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ ($\gtrsim 10^{-12}\,{\rm g\,cm^{-3}\,day}$), the numerical results are broadly consistent with a linear relation between $r_{\rm gr,final}$ and $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$. At small $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ ($\sim 10^{-16}\,{\rm g\,cm^{-3}\,day}$), the grain size approaches the monomer size, indicating that substantial growth

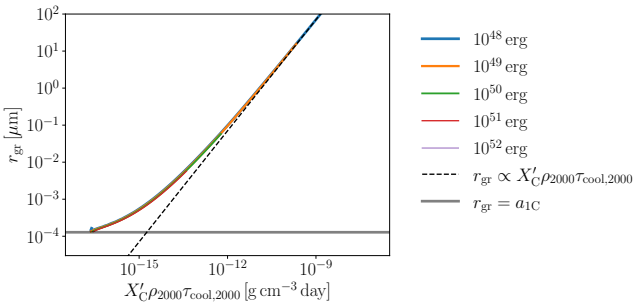


Fig. 4. Final mass-averaged graphite grain radius as a function of $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ for Ni-free adiabatic models with $R_0 = 10^{13}$ cm. Here, ρ_{2000} and $\tau_{\rm cool,2000}$ are the gas density and cooling timescale evaluated when the gas temperature reaches $T = 2000$ K, and $X'_{\rm C}$ is the effective carbon abundance available for graphite formation. Different colored curves correspond to different explosion energies. The black dashed line shows a slope-one guide corresponding to $r_{\rm gr} \propto X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$. The gray horizontal line indicates the monomer radius. The same collapse is obtained for models with $R_0 = 10^{11}$–10^{14} cm.

does not occur. In the present calculations, a graphite grain size of $r_{\rm gr,final} \simeq 1\,\mu$m corresponds approximately to

$$X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000} \simeq 1.2\times10^{-11}\,{\rm g\,cm^{-3}\,day}. \tag{34}$$

This value is an empirical threshold obtained from the present graphite-growth calculations rather than a first-principles analytic boundary. Because this relation is expressed entirely in terms of local quantities during the graphite-growth phase, it does not explicitly depend on the equation of state. The equation of state enters when ρ_{2000} and $\tau_{\rm cool,2000}$ are mapped onto the macroscopic ejecta parameters, such as $E_{\rm exp}$, $M_{\rm ej}$, and R_0. We examine this mapping in the next subsection.

4.3 Dependence of graphite grain size on supernova parameters in Ni-free ejecta

We now interpret the parameter dependence shown in Fig. 3 using the grain-growth estimate developed in Section 4.2. Specifically, we derive how $X'_C\rho_{2000}\tau_{\rm cool,2000}$ depends on $E_{\rm exp}$, $M_{\rm ej}$, and R_0 in the matter-dominated adiabatic limit. Mapping this quantity onto $E_{\rm exp}$, $M_{\rm ej}$, and R_0, however, depends on the thermal evolution of the ejecta. For adiabatic expansion with an approximately constant adiabatic index, the temperature and density satisfy $T\propto\rho^{\gamma-1}$. Thus, in the radiation- and matter-dominated limits,

$$T \propto \begin{cases} \rho^{1/3} & \text{(radiation dominated)},\\ \rho^{2/3} & \text{(matter dominated)}. \end{cases} \tag{35}$$

Since graphite growth occurs around $T \simeq 2000$ K, the density exposure at 2000 K is important for grain growth. For a given decrease in temperature, radiation-dominated ejecta experience stronger density dilution than matter-dominated ejecta and are therefore less favorable for forming large grains. Here, we focus on the matter to matter adiabatic case, for which a simple analytic scaling can be derived.

For the matter to matter case, as shown in Appendix 1, the density and cooling timescale at $T = 2000$ K scale as $\rho_{2000} \propto R_0^{-3}M_{\rm ej}^{5/2}E_{\rm exp}^{-3/2}$ and $\tau_{\rm cool,2000}\propto R_0$, respectively. Eq. (33) identifies $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ as the local combination that controls grain growth (a physical motivation for this combination, based on the timescale ratio of Yamamoto & Hasegawa (1977), is given in Section 5.2). Here, restricting ourselves to the efficient-growth branch in the matter-dominated adiabatic limit, we relate this local quantity to the macroscopic ejecta parameters $E_{\rm exp}$, $M_{\rm ej}$, and R_0. Therefore, $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ scales as

$$X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000} \propto X'_{\rm C}E_{\rm exp}^{-3/2}M_{\rm ej}^{5/2}R_0^{-2}. \tag{36}$$

This scaling reproduces the numerical trend that grain growth is favored by lower $E_{\rm exp}$, larger $M_{\rm ej}$, and smaller R_0. The analytic argument determines the parameter dependence but not its normalization. Using the numerical normalization of the efficient-growth branch in Fig. 4, we obtain

$$r_{\rm gr,final} \simeq 615\,\mu{\rm m}\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{-3/2}\left(\frac{M_{\rm ej}}{5\,M_\odot}\right)^{5/2} \times\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{-2}\left(\frac{X'_{\rm C}}{3\times10^{-3}}\right). \tag{37}$$

Eq. (37) should be interpreted as a numerical fit to the efficient-growth branch in the matter-dominated adiabatic limit, not as the radius obtained by allowing grains to grow over the full local cooling time.

For reference, setting $r_{\rm gr,final} = 1\,\mu$m in Eq. (37) gives the formal micron-size contour,

$$M_{\rm ej,1\mu m}^{\rm mat} \simeq 0.38 M_\odot\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{3/5}\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{4/5} \times\left(\frac{X'_{\rm C}}{3\times10^{-3}}\right)^{-2/5}. \tag{38}$$

The formal contour in Eq. (38) is meaningful only where the ejecta is actually matter dominated during the cooling from the initial state to the graphite-condensation phase.

A useful self-consistency condition for the matter-dominated branch is obtained from the ratio of the initial radiation and gas internal energies,

$$\eta_0 := \frac{\varepsilon_{\rm rad,0}}{\varepsilon_{\rm mat,0}} \simeq 1.0\times10^{-2}\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{3}\left(\frac{M_{\rm ej}}{5\,M_\odot}\right)^{-4} \times\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{3}\left(\frac{f_{\rm int}}{0.9}\right)^{3}. \tag{39}$$

The matter-dominated scaling is self-consistent only for $\eta_0 \lesssim 1$, or equivalently

$$M_{\rm ej} \gtrsim 1.6\,M_\odot\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{3/4}\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{3/4}\left(\frac{f_{\rm int}}{0.9}\right)^{3/4}. \tag{40}$$

Comparing the four panels at fixed $E_{\rm exp}$ and $M_{\rm ej}$ shows that the final grain size systematically decreases as R_0 increases. In particular, the $1\,\mu$m contour shifts toward lower $E_{\rm exp}$ and higher $M_{\rm ej}$ with increasing R_0. Thus, the analytic scaling provides a physical interpretation of the numerical trends in Fig. 3, while the full one-zone calculations determine the grain sizes outside the self-consistent matter-dominated regime. These results show that, in the Ni-free adiabatic limit, low-energy and massive ejecta provide the most favorable hydrodynamical conditions for forming micron-sized graphite grains. We next examine how this picture is modified by radioactive heating and radiative cooling.

4.4 The effect of Ni heating

In Section 4.3, we considered Ni-free adiabatic models with a fixed effective carbon abundance of $X'_{\rm C} = 3\times10^{-3}$. We now include radioactive heating and radiative cooling, and examine how the grain size depends on $E_{\rm exp}$, $M_{\rm Ni}$, and $X'_{\rm C}$, while fixing $R_0 = 10^{12}$ cm, $M_{\rm ej} = 10\,M_\odot$, and $f_{\rm int} = 0.9$. Figure 5 shows the

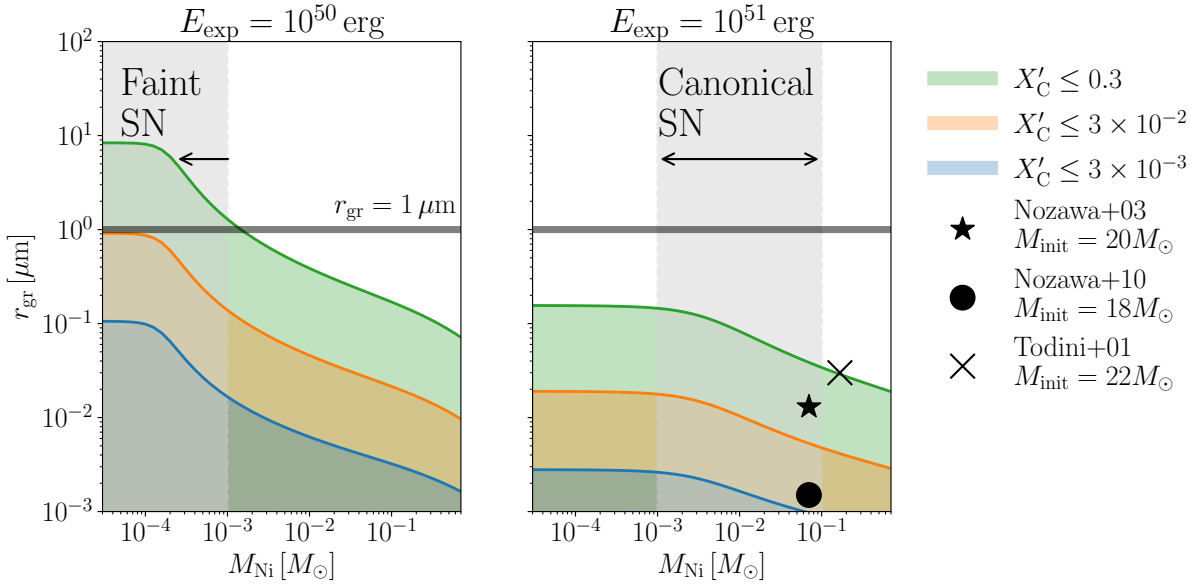


Fig. 5. The graphite grain sizes ($r_{\rm gr}$) as a function of the ejected ^{56}Ni mass ($M_{\rm Ni}$). The left and right panels show models with $E_{\rm exp} = 10^{50}\,{\rm erg}$ and $E_{\rm exp} = 10^{51}\,{\rm erg}$, respectively. The green, orange, and blue curves correspond to $X'_{\rm C} = 3\times10^{-1}$, 3×10^{-2}, and 3×10^{-3}, respectively, while the corresponding shaded regions show the ranges $X'_{\rm C} \le 3\times10^{-1}$, $X'_{\rm C} \le 3\times10^{-2}$, and $X'_{\rm C} \le 3\times10^{-3}$. The gray shaded regions are labeled "Faint SN" in the left panel and "Canonical SN" in the right panel. The horizontal gray line marks $r_{\rm gr} = 1\,\mu{\rm m}$. The symbols show representative graphite grain sizes from previous supernova dust-formation calculations by Nozawa et al. (2003), Nozawa et al. (2010), and Todini & Ferrara (2001), for the progenitor masses indicated in the legend.

representative graphite grain size as a function of $M_{\rm Ni}$. The left and right panels show models with $E_{\rm exp} = 10^{50}\,{\rm erg}$ and $10^{51}\,{\rm erg}$, respectively. The blue, orange, and green curves correspond to $X'_{\rm C} = 3\times10^{-3}$, 3×10^{-2}, and 3×10^{-1}, while the shaded regions indicate the ranges of $X'_{\rm C}$ adopted for the three illustrative composition cases. These three values are chosen as illustrative benchmarks spanning the carbon abundances from the H-rich envelope to He-rich and carbon–oxygen-rich layers of massive-star progenitors (Umeda et al. 2012; Asplund et al. 2021). The gray shaded regions indicate the faint-SN ($M_{\rm Ni} < 10^{-3}\,M_\odot$) and canonical-SN ($10^{-3} \le M_{\rm Ni}/M_\odot \le 10^{-1}$) regimes adopted here, motivated by the ^{56}Ni mass ranges discussed in Section 1. The symbols show grain sizes from Nozawa et al. (2003) and Nozawa et al. (2010) for progenitor masses of $20\,M_\odot$ and $18\,M_\odot$, respectively. The cross shows the result for the $22\,M_\odot$ model of Todini & Ferrara (2001).

Fig. 5 shows three systematic trends. At fixed $E_{\rm exp}$ and $M_{\rm Ni}$, the final grain size increases with increasing $X'_{\rm C}$. At fixed $M_{\rm Ni}$ and $X'_{\rm C}$, lower explosion energy produces larger grains. For both explosion energies, increasing $M_{\rm Ni}$ suppresses grain growth, while the dependence on $M_{\rm Ni}$ becomes weak toward the low-$M_{\rm Ni}$ regime. At sufficiently small $M_{\rm Ni}$, radioactive heating becomes subdominant, and the thermal evolution is governed mainly by adiabatic expansion and radiative cooling. In the limiting case $M_{\rm Ni} = 0$, the resulting grain sizes approach those obtained in the Ni-free adiabatic calculations discussed in the previous subsection.

We can understand these features as follows. In Eq. (33), the final grain size is set by the product $\rho_{2000}\tau_{\rm cool,2000}$. Radioactive heating lengthens $\tau_{\rm cool,2000}$, but it also delays condensation while the ejecta continue to expand, lowering ρ_{2000} by orders of magnitude (Fig. 1). The latter dominates, and the resulting decrease in the product reduces $r_{\rm gr,final}$.

For $E_{\rm exp} = 10^{51}\,{\rm erg}$, micron-sized graphite does not form over the explored range of $X'_{\rm C}$. Within the canonical-SN regime, even the largest adopted carbon abundance gives $r_{\rm gr} \lesssim 0.1$–$0.2\,\mu{\rm m}$, with progressively smaller grains toward lower $X'_{\rm C}$. By contrast, in the faint-SN regime with $E_{\rm exp} = 10^{50}\,{\rm erg}$, graphite grains with radii of order 1–10 μm can form for $X'_{\rm C} \sim 3\times10^{-2}$–$3\times10^{-1}$, representative of the He layer and carbon–oxygen core.

5 Discussion

5.1 Implications for the origin of presolar graphite

Our results identify low-energy, nickel-poor explosions as possible sources of micron-sized presolar graphite. In the models shown in Fig. 5, with $M_{\rm ej} = 10\,M_\odot$, $R_0 = 10^{12}\,{\rm cm}$, and $f_{\rm int} = 0.9$, explosions with $E_{\rm exp} = 10^{50}\,{\rm erg}$ and $M_{\rm Ni} \lesssim 10^{-3}\,M_\odot$ can produce graphite grains with sizes of order 1–10 μm for $X'_{\rm C} \sim 0.03$–0.3. These sizes are comparable to those of large presolar graphite grains identified as supernova condensates (Amari et al. 2014). Thus, the discrepancy between measured presolar grain sizes and previous calculations may partly reflect the range of explosion parameters considered.

Observationally inferred explosion parameters indicate that such low-energy, nickel-poor explosions are not rare. Martinez et al. (2022b) obtained median explosion energies of $6.1\times10^{50}\,{\rm erg}$ for their gold sample of type II supernovae and $6.3\times10^{50}\,{\rm erg}$ for their full sample, below the commonly adopted $10^{51}\,{\rm erg}$. Nickel masses also span a broad distribution. In the type II supernova comparison sample used by Meza & Anderson (2020), more than 25% have $M_{\rm Ni} < 0.015\,M_\odot$. Furthermore, Müller et al. (2017) found a positive correlation between explosion energy and nickel mass. Lower-energy explosions therefore tend to combine two properties that favor grain growth in our models. These observations motivate examining the low-energy, low-nickel part of the population as a source of large graphite grains. However, they do

not establish the fraction of supernovae that satisfy the micron-grain formation conditions identified here.

The ejecta mass is also constrained observationally. In the sample of Martinez et al. (2022b), $M_{\rm ej}$ is concentrated near 8–$10\,M_\odot$, with a median of $8.4\,M_\odot$ for the gold sample, and is only weakly correlated with the explosion energy (Pearson coefficient 0.36 ± 0.11; Martinez et al. 2022a). The observed type II supernovae therefore occupy a band of $M_{\rm ej} \sim 10\,M_\odot$ and $E_{\rm exp} \gtrsim 10^{50}\,{\rm erg}$ in Fig. 3; the upper-left region of the figure, $E_{\rm exp} \sim 10^{48}\,{\rm erg}$ with $M_{\rm ej} \sim 10\,M_\odot$, where our models yield the largest grains, has no observed supernova counterpart.

The contribution of these grains to planetary systems also depends on their survival and transport. Grain-size-dependent shock processing can modify the population that survives after formation (Nozawa et al. 2006; Nozawa et al. 2007). Grain size also affects injection into protoplanetary disks. The short-lived radionuclide enrichment model of Sawada et al. (2024) includes direct injection by supernova dust grains with radii $\gtrsim 1\,\mu{\rm m}$, which enter the disk more efficiently than small grains. Our calculations show that graphite can reach this size under favorable explosion conditions. They therefore motivate accounting for the dependence of grain formation on explosion parameters when modeling solid delivery to disks. Quantifying radionuclide delivery additionally requires the incorporation of the relevant nuclides into dust, together with grain survival and transport.

A further test is whether the same ejecta conditions can reproduce both the grain sizes and isotopic compositions. Measurements of C, O, Si, and Ti isotopes in individual graphite grains provide constraints on their source material and mixing between ejecta layers (Travaglio et al. 1999; Jadhav et al. 2013). In particular, Schofield et al. (2022) found that their low- and intermediate-energy models without a C/Si zone require mixing with oxygen-rich ejecta to explain the ^{28}Si and ^{44}Ti signatures. Testing the low-energy origin proposed here therefore requires coupling grain formation to nucleosynthesis and ejecta mixing.

5.2 Comparison with Yamamoto and Hasegawa (1977)

The quantity $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ examined in this work revisits the timescale ratio $\Lambda_{\rm on}$ introduced by Yamamoto & Hasegawa (1977) and applied to supernova ejecta by Nozawa & Kozasa (2013). Yamamoto & Hasegawa (1977) characterized grain formation using the ratio of the supersaturation timescale to the monomer collision timescale. Following Nozawa & Kozasa (2013), we consider this ratio at the onset of dust formation, $t_{\rm on}$, before appreciable monomer depletion. For graphite, the undepleted carbon number density is $\tilde{c}_{\rm 1C} = X'_{\rm C}\rho/m_{\rm 1C}$, and the timescale ratio can be written as

$$\Lambda_{\rm on,C} \equiv \frac{\tau_{\rm sat,C}(t_{\rm on})}{\tau_{\rm coll,C}(t_{\rm on})} = \left.\frac{4\pi\alpha_{\rm C}a_{\rm 1C}^2}{m_{\rm 1C}}\frac{v_{\rm th,C}}{|d\ln S_{\rm C}/d\ln T|}X'_{\rm C}\rho\tau_{\rm cool}\right|_{t=t_{\rm on}}. \tag{41}$$

Here, $\tau_{\rm sat,C} \equiv |d\ln S_{\rm C}/dt|^{-1}$, $\tau_{\rm coll,C} = (4\pi\alpha_{\rm C}a_{\rm 1C}^2\tilde{c}_{\rm 1C}v_{\rm th,C})^{-1}$, and $\tau_{\rm cool} \equiv |d\ln T/dt|^{-1}$.

Before appreciable depletion, $|d\ln S_{\rm C}/d\ln T| \simeq A/T \simeq 40$ at $T = 2000\,{\rm K}$ for adiabatic cooling (Appendix 2), so that $\tau_{\rm sat,C} \simeq (T/A)\tau_{\rm cool}$, as in the approximation used by Nozawa & Kozasa (2013), and Eq. (41) reduces to $\Lambda_{\rm on,C} \propto X'_{\rm C}\rho\tau_{\rm cool}$. This provides the physical motivation for our use of $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$, with 2000 K adopted as a common reference temperature, whereas $\Lambda_{\rm on,C}$ is evaluated at the actual onset of dust formation. Expressed in this way, the timescale ratio is no longer tied to a particular ejecta model, but can be mapped directly onto the macroscopic explosion parameters, as was done in Section 4.3 to obtain the scaling of Eq. (36) with $E_{\rm exp}$, $M_{\rm ej}$, and R_0.

5.3 Caveat

The present one-zone calculations identify conditions favorable for graphite formation, but do not resolve the density and composition structure, clumping, or mixing within the ejecta. The adopted carbon abundances therefore represent illustrative compositions rather than individual ejecta layers with distinct thermal histories. In particular, the absence of micron-sized grains in our canonical-energy models does not exclude their formation in locally dense, carbon-rich regions. Conversely, the favorable conditions found at low explosion energies require that carbon-rich material be ejected.

Whether such material escapes the progenitor depends on fallback and on mixing within the ejecta. In one-dimensional models without mixing, carbon–oxygen core material is ejected only when $E_{\rm exp}$ exceeds the binding energy of the hydrogen envelope (Shinoda et al. 2026). Even above this threshold, the amount of helium-layer or carbon–oxygen core material that escapes depends on mixing, which has been parameterized in one-dimensional models as the mixing-fallback framework (Umeda & Nomoto 2002; Tominaga et al. 2007); a quantitative treatment requires multi-dimensional calculations (Tominaga 2009; Chan et al. 2020).

The available carbon abundance $X'_{\rm C}$ is prescribed, without explicitly following the formation and destruction of CO or other molecules. The effects of radioactive heating are thus evaluated at a specified carbon availability, while radioactive decay may also affect the chemistry that determines this availability. The thermal evolution is likewise approximate, with a fixed gas equation of state, constant opacities, and simplified radiative cooling and radioactive energy deposition. These assumptions introduce uncertainties in the temperature history and hence in the density and cooling timescale during grain formation.

The dust calculation adopts steady-state classical nucleation theory, bulk material properties for small clusters, a unit sticking probability, and equal gas and condensed-phase temperatures. The sensitivity of the predicted grain sizes to these assumptions, including departures from steady-state nucleation, remains to be assessed. Finally, we calculate grain sizes at formation and do not follow subsequent processing by reverse shocks or destruction in the interstellar medium. Connecting these results to presolar graphite therefore requires both a viable formation environment and the survival of the resulting grains.

6 Conclusions

We investigated graphite formation in supernova ejecta by combining a one-zone model of the ejecta evolution with calculations of nucleation and grain growth. We examined how the explosion parameters, radioactive heating, and available carbon abundance affect the formation of micron-sized graphite.

For the Ni-free adiabatic models, the final grain radius approximately follows a single relation with $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ over the explored parameter range. This numerical relation connects the final grain size to the carbon supply and cooling timescale during the grain-growth phase. In the efficient-growth regime, the grain radius is approximately proportional to this product. For ejecta that remain matter dominated during adiabatic cooling, the prod-

uct scales as $X'_{\rm C}E_{\rm exp}^{-3/2}M_{\rm ej}^{5/2}R_0^{-2}$, linking efficient grain growth to lower explosion energies, larger ejecta masses, and smaller initial radius within this regime.

Our results identify low-energy explosions with small ejected ^{56}Ni masses as promising environments for forming large graphite grains, provided that sufficient carbon-rich material is ejected. The conditions identified here provide a basis for testing a faint-supernova origin of large presolar graphite with models that resolve the ejecta structure and chemistry and follow the subsequent survival of the grains.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

Appendix 1 Analytic scaling in the matter-dominated adiabatic limit

We derive the scalings used in Section 4.3 for Ni-free, matter-dominated adiabatic ejecta with $\gamma = 5/3$. For the analytic estimate we adopt a coasting expansion,

$$R(t) = R_0(1 + t/t_0), \qquad t_0 = R_0/v, \tag{A1}$$

$$v = \left(\frac{2E_{\rm exp}}{M_{\rm ej}}\right)^{1/2}. \tag{A2}$$

The gas-only initial temperature and density are

$$T_0 = \frac{(\gamma-1)m_p}{k_B}\frac{f_{\rm int}E_{\rm exp}}{M_{\rm ej}}, \tag{A3}$$

$$\rho_0 = \frac{3M_{\rm ej}}{4\pi R_0^3}. \tag{A4}$$

Mass conservation and $T \propto \rho^{\gamma-1}$ give $\rho(t)/\rho_0 = (1 + t/t_0)^{-3}, T(t)/T_0 = (1 + t/t_0)^{-2}$. At any reference temperature $T_{\rm r} \le T_0$, these relations imply

$$1 + \frac{t_{\rm r}}{t_0} = \left(\frac{T_0}{T_{\rm r}}\right)^{1/2}, \qquad t_{\rm r} = t_0\left[\left(\frac{T_0}{T_{\rm r}}\right)^{1/2} - 1\right]. \tag{A5}$$

Differentiating the temperature history gives

$$\frac{d\ln T}{dt} = -\frac{2}{t+t_0}, \qquad \tau_{\rm cool}(t) \equiv \left|\frac{d\ln T}{dt}\right|^{-1} = \frac{t+t_0}{2}. \tag{A6}$$

Consequently,

$$\tau_{\rm cool,r} = \frac{R_0}{2v}\left(\frac{T_0}{T_{\rm r}}\right)^{1/2} = 7.80\,{\rm day}\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)\left(\frac{f_{\rm int}}{0.9}\right)^{1/2}\left(\frac{T_{\rm r}}{2000\,{\rm K}}\right)^{-1/2}. \tag{A7}$$

The factors $(M_{\rm ej}/E_{\rm exp})^{1/2}$ from v^{-1} and $(E_{\rm exp}/M_{\rm ej})^{1/2}$ from $T_0^{1/2}$ cancel. This cancellation assumes fixed $f_{\rm int}$. The corresponding density is

$$\rho_{\rm r} = \frac{3M_{\rm ej}}{4\pi R_0^3}\left[\frac{(\gamma-1)m_p f_{\rm int}E_{\rm exp}}{k_B T_{\rm r} M_{\rm ej}}\right]^{-3/2} = 3.40\times10^{-7}\,{\rm g\,cm^{-3}}\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{-3/2}\left(\frac{M_{\rm ej}}{5M_\odot}\right)^{5/2} \times\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{-3}\left(\frac{f_{\rm int}}{0.9}\right)^{-3/2}\left(\frac{T_{\rm r}}{2000\,{\rm K}}\right)^{3/2}. \tag{A8}$$

Multiplication of Eqs. (A7) and (A8) yields

$$X'_{\rm C}\rho_{\rm r}\tau_{\rm cool,r} = 7.95\times10^{-9}\,{\rm g\,cm^{-3}\,day}\left(\frac{X'_{\rm C}}{3\times10^{-3}}\right) \times\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{-3/2}\left(\frac{M_{\rm ej}}{5M_\odot}\right)^{5/2}\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{-2} \times\left(\frac{f_{\rm int}}{0.9}\right)^{-1}\left(\frac{T_{\rm r}}{2000\,{\rm K}}\right). \tag{A9}$$

For $T_{\rm r} = 2000\,{\rm K}$, this is the exposure used in the main text. If the effective growth factor and microscopic properties are fixed, Eq. (33) then gives

$$r_{\rm gr,final} - a_{\rm 1C} \propto X'_{\rm C}f_{\rm int}^{-1}E_{\rm exp}^{-3/2}M_{\rm ej}^{5/2}R_0^{-2}. \tag{A10}$$

At fixed R_0, $X'_{\rm C}$, and $f_{\rm int}$, constant-size contours therefore satisfy $M_{\rm ej} \propto E_{\rm exp}^{3/5}$. Here the effective growth factor is assumed to be approximately constant.

To test the gas-only approximation, we evaluate the initial ratio of radiation to gas energy using the gas-only T_0,

$$\eta_0 = \frac{2am_p}{3k_B}\frac{T_0^3}{\rho_0} \tag{A11}$$

Along the gas adiabat, $\eta(T) = \eta_0(T/T_0)^{3/2}$, so radiation becomes less important as the ejecta cools. The gas-dominated limit requires $\eta_0 \ll 1$; the approximate transition $\eta_0 \lesssim 1$ corresponds to

$$M_{\rm ej} \gtrsim 1.58M_\odot\left(\frac{E_{\rm exp}}{10^{48}\,{\rm erg}}\right)^{3/4}\left(\frac{R_0}{10^{12}\,{\rm cm}}\right)^{3/4} \times\left(\frac{f_{\rm int}}{0.9}\right)^{3/4}. \tag{A12}$$

Near this transition the full gas-plus-radiation equation of state is needed for quantitative results.

Appendix 2 Collision exposure and effective growth window

Here we give a physical interpretation of the product $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ used in Section 4.2. The quantity $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$ measures the potential exposure of graphite grains to carbon monomers over the local cooling time around $T = 2000\,{\rm K}$. Let n_1 be the number density of carbon key species, $n_{\rm gr}$ the grain number density. We use the same flux speed as in the main text, $v_{\rm th} = \sqrt{k_BT/(2\pi m_{\rm 1C})}$.

If grains grew over the full local cooling time, the size increment would be proportional to $n_{1,2000}v_{\rm th,2000}\tau_{\rm cool,2000}$, or equivalently to $X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}$. However, this overestimates the actual collision exposure because stable grains exist and grow only during a short interval around the nucleation burst. Before this burst, the number of stable grains is negligible. After the burst, the rapid increase in the total grain surface area, $4\pi n_{\rm gr}r_{\rm gr}^2$, shortens the depletion timescale and reduces the supersaturation ratio,

thereby quenching both further nucleation and growth.

Thus, the effective collision-active duration is not the full cooling time but

$$\Delta t_{\rm coll} = f_{\rm grow}\tau_{\rm cool,2000}, \tag{A13}$$

where $f_{\rm grow} \ll 1$ represents the fraction of the cooling time during which monomer–grain collisions effectively contribute to grain growth.

The smallness of $f_{\rm grow}$ is expected from the sharp temperature dependence of the steady-state nucleation rate. The nucleation burst opens the growth window, while monomer depletion closes it. If depletion closes the growth window on a comparable timescale, the effective collision duration can be estimated from the temperature width over which nucleation becomes efficient,

$$f_{\rm grow} \sim \frac{\Delta t_{\rm coll}}{\tau_{\rm cool}} \sim \Delta \ln T_{\rm nuc}. \tag{A14}$$

Because the nucleation rate depends exponentially on the supersaturation ratio, this temperature width is much smaller than unity in logarithmic temperature.

We now estimate this temperature width from the strongly temperature-sensitive exponential factor of the steady-state nucleation rate. Ignoring slowly varying prefactors, we write

$$J \propto \exp\left(-\frac{\mathcal{B}_{\rm nuc}}{s^2}\right), \tag{A15}$$

$$s \equiv \ln S, \qquad \mathcal{B}_{\rm nuc} \equiv \frac{4\mu^3}{27}. \tag{A16}$$

Here S is the supersaturation ratio and μ is the dimensionless surface-energy parameter. For graphite,

$$\mu = \frac{4\pi a_{1\rm C}^2 \zeta_{\rm C}}{k_B T} \simeq 10.45 \left(\frac{a_{1\rm C}}{1.281\times10^{-8}\,{\rm cm}}\right)^2 \left(\frac{\zeta_{\rm C}}{1400\,{\rm erg\,cm^{-2}}}\right) \left(\frac{T}{2000\,{\rm K}}\right)^{-1}. \tag{A17}$$

The logarithmic variation over a small temperature interval is $|\Delta \ln J| \simeq |d\ln J/d\ln T|\,|\Delta \ln T|$. Using $\mathcal{B}_{\rm nuc} \propto T^{-3}$ gives

$$\frac{d\ln J}{d\ln T} \simeq \frac{3\mathcal{B}_{\rm nuc}}{s^2} + \frac{2\mathcal{B}_{\rm nuc}}{s^3}\frac{ds}{d\ln T}. \tag{A18}$$

Around the nucleation epoch, s changes rapidly because it contains both the equilibrium and carbon partial-pressure terms. Before depletion on a matter-dominated adiabat, $q \equiv -ds/d\ln T \simeq A/T - 5/2 \simeq 40.7$ at 2000 K. At $(\mu, s, q) = (10, 4, 40)$, the slope magnitude is approximately 157. The corresponding local one-e-fold temperature interval is

$$\Delta \ln T_{\rm nuc} \sim \left|\frac{d\ln J}{d\ln T}\right|^{-1} \simeq 6.35\times10^{-3}\left(\frac{\mu}{10}\right)^{-3}\left(\frac{s}{4}\right)^{3}\left(\frac{q}{40}\right)^{-1} \times \left[\frac{|1-3s/(2q)|}{0.85}\right]^{-1}. \tag{A19}$$

This estimate applies on the rising side of the burst, where the slope varies slowly over the interval, rather than at the maximum where the first derivative vanishes.

Since $\tau_{\rm cool} = |d\ln T/dt|^{-1}$, the associated interval is

$$\Delta t_{\rm coll} \simeq 0.635\,{\rm day}\left(\frac{\tau_{\rm cool}}{100\,{\rm day}}\right)\left(\frac{\Delta \ln T_{\rm nuc}}{6.35\times10^{-3}}\right). \tag{A20}$$

Under the short nucleation-and-depletion window approximation above,

$$f_{\rm grow} \simeq 6.35\times10^{-3}\left(\frac{\mu}{10}\right)^{-3}\left(\frac{s}{4}\right)^{3}\left(\frac{q}{40}\right)^{-1} \times \left[\frac{|1-3s/(2q)|}{0.85}\right]^{-1} \tag{A21}$$

This is an order-of-magnitude estimate of the effective interval, not an identification of the microscopic growth and nucleation timescales.

This estimate can be compared with the normalization of the empirical large-exposure relation in the main text,

$$r_{\rm gr} \simeq 0.68\,\mu{\rm m}\left(\frac{X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}}{10^{-11}\,{\rm g\,cm^{-3}\,day}}\right). \tag{A22}$$

On the other hand, using the full cooling time as the collision-active duration at $T = 2000\,{\rm K}$ gives

$$r_{\rm gr} - a_{1\rm C} \simeq 1.77\times10^{2}\,\mu{\rm m}\,\alpha_{\rm C}\left(\frac{a_{1\rm C}}{1.281\times10^{-8}\,{\rm cm}}\right)^3 \times \left(\frac{X'_{\rm C}\rho_{2000}\tau_{\rm cool,2000}}{10^{-11}\,{\rm g\,cm^{-3}\,day}}\right). \tag{A23}$$

For grains much larger than the monomer, the ratio of these normalizations gives

$$f_{\rm grow,eff} \simeq 3.83\times10^{-3}\,\alpha_{\rm C}^{-1}\left(\frac{a_{1\rm C}}{1.281\times10^{-8}\,{\rm cm}}\right)^{-3}. \tag{A24}$$

This value is comparable to the nucleation-window estimate. The smaller normalization of the numerical grain-size relation can therefore be interpreted in terms of a short effective collision interval around nucleation and depletion, rather than the full cooling time.

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